

求下面各机械臂的 DH 参数、齐次转换矩阵和雅可比矩阵

1.

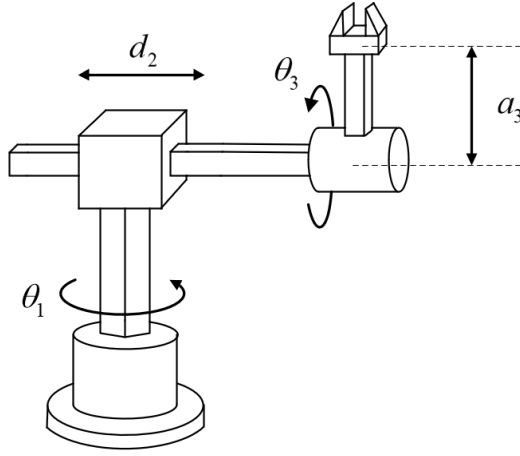


图 1

DH-table

α_{i-1}	a_{i-1}	d_i	θ_i
0	0	0	θ_1
-90	0	d_2	0
0	0	0	θ_3

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad 3'$$

$${}^1T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad 3'$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad 3'$$

$${}^3T_e = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad 1'$$

$${}^0T_e = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_e \quad 2'$$

$$= \begin{bmatrix} c_1c_3 & -s_1 & c_1s_3 & -d_2s_1 - ac_1s_3 \\ c_3s_1 & c_1 & s_1s_3 & d_2c_1 - as_1s_3 \\ -s_3 & 0 & c_3 & -ac_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad 3'$$

$$P = \begin{bmatrix} -d_2s_1 - ac_1s_3 \\ d_2c_1 - as_1s_3 \\ -ac_3 \end{bmatrix} \quad 2'$$

$$Z_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, Z_2 = \begin{pmatrix} -s_1 \\ c_1 \\ 0 \end{pmatrix}, Z_3 = \begin{pmatrix} -s_1 \\ c_1 \\ 0 \end{pmatrix} \quad 3'$$

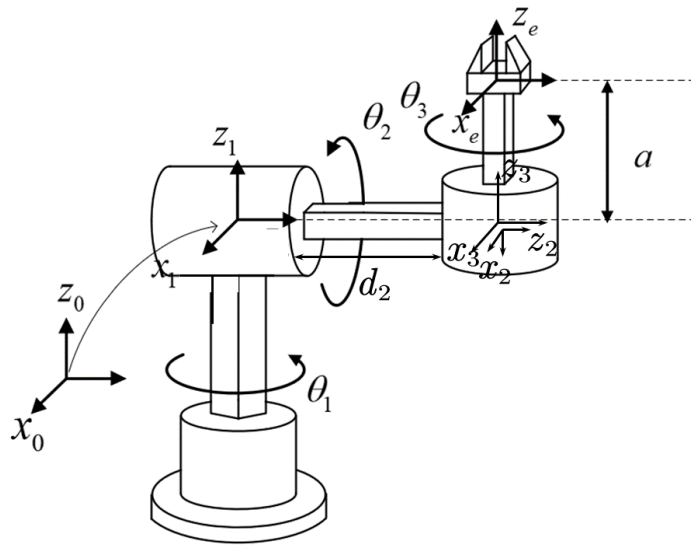
$$\xi_1 = 1, \xi_2 = 0, \xi_3 = 1, \quad 3'$$

$$\theta = [\theta_1 \quad d_2 \quad \theta_3]^T$$

$$J(\theta) = \begin{bmatrix} \frac{\partial P_x}{\partial \theta_1} & \frac{\partial P_x}{\partial d_2} & \frac{\partial P_x}{\partial \theta_3} \\ \frac{\partial P_y}{\partial \theta_1} & \frac{\partial P_y}{\partial d_2} & \frac{\partial P_y}{\partial \theta_3} \\ \frac{\partial P_z}{\partial \theta_1} & \frac{\partial P_z}{\partial d_2} & \frac{\partial P_z}{\partial \theta_3} \\ \xi_1 Z_1 & \xi_2 Z_2 & \xi_3 Z_3 \end{bmatrix} \quad 4'$$

$$= \begin{bmatrix} -d_2c_1 + as_1s_3 & -s_1 & -ac_1c_3 \\ -d_2s_1 - ac_1s_3 & c_1 & -as_1c_3 \\ 0 & 0 & as_3 \\ 0 & 0 & -s_1 \\ 0 & 0 & c_1 \\ 1 & 0 & 0 \end{bmatrix} \quad 8'$$

2.



DH-table

α_{i-1}	a_{i-1}	d_i	θ_i
0	0	0	θ_1
-90	0	d_2	θ_2
90	0	0	θ_3
0	0	0	a

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^3'$$

$${}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^3'$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^3'$$

$${}^3T_e = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^1'$$

$${}^0T_e = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_e {}^2'$$

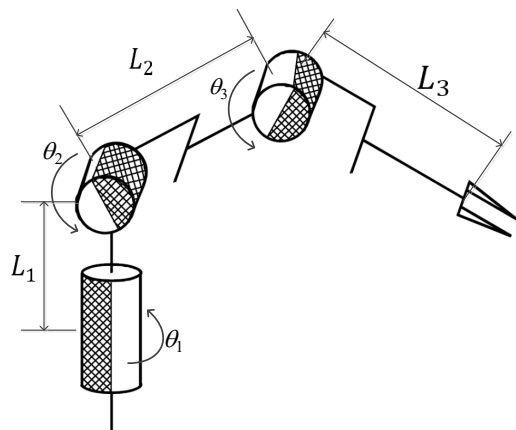
$$\begin{aligned}
&= \begin{bmatrix} c_1 c_2 c_3 - s_1 s_3 & -c_3 s_1 - c_1 c_2 s_3 & c_1 s_2 & -s_1 d_2 + a c_1 s_2 \\ c_1 s_3 + c_2 c_3 s_1 & c_1 c_3 - c_2 s_1 s_3 & s_1 s_2 & c_1 d_2 + a s_1 s_2 \\ -c_3 s_2 & s_2 s_3 & c_2 & a c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} 3', \\
P &= \begin{bmatrix} a c_1 s_2 \\ a s_1 s_2 \\ a c_2 \end{bmatrix} 2', \\
Z_1 &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, Z_2 = R_x(\theta_1) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -s_1 \\ c_1 \\ 0 \end{pmatrix}, Z_3 \\
&= R_x(\theta_1) R_y(\theta_2) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & 0 & s_2 \\ 0 & 1 & 1 \\ -s_2 & 0 & c_2 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{pmatrix} 3'
\end{aligned}$$

$$\xi_1 = 1, \xi_2 = 1, \xi_3 = 1, 3'$$

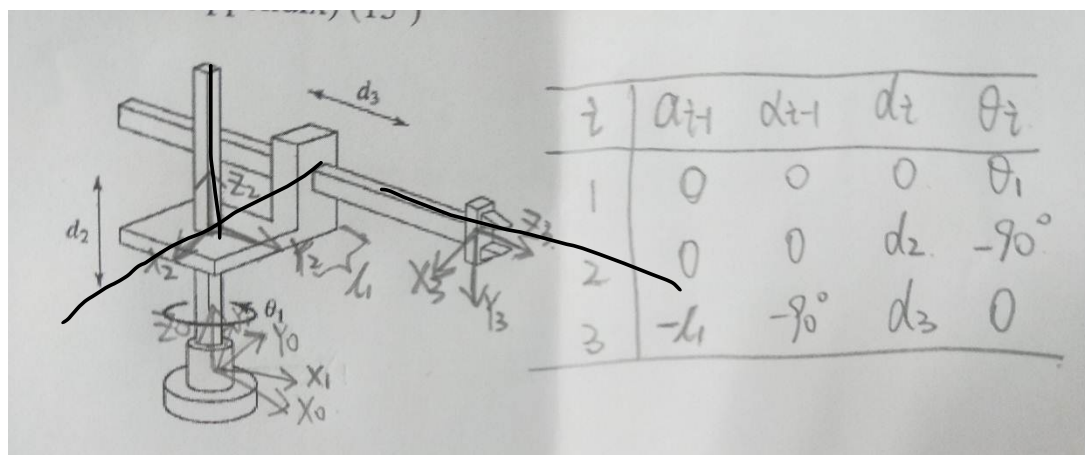
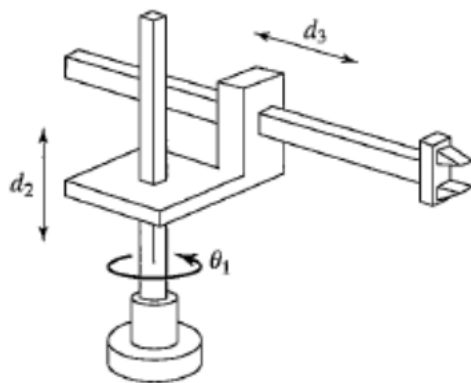
$$\theta = [\theta_1 \quad \theta_2 \quad \theta_3]^T$$

$$\begin{aligned}
J(\theta) &= \begin{bmatrix} \frac{\partial P_x}{\partial \theta_1} & \frac{\partial P_x}{\partial \theta_2} & \frac{\partial P_x}{\partial \theta_3} \\ \frac{\partial P_y}{\partial \theta_1} & \frac{\partial P_y}{\partial \theta_2} & \frac{\partial P_y}{\partial \theta_3} \\ \frac{\partial P_z}{\partial \theta_1} & \frac{\partial P_z}{\partial \theta_2} & \frac{\partial P_z}{\partial \theta_3} \\ \xi_1 Z_1 & \xi_2 Z_2 & \xi_3 Z_3 \end{bmatrix} 4', \\
&= \begin{bmatrix} -c_1 d_2 - a s_1 s_2 & a c_1 c_2 & 0 \\ -s_1 d_2 + a c_1 s_2 & a s_1 c_2 & 0 \\ 0 & -a s_2 & 0 \\ 0 & -s_1 & c_1 s_2 \\ 0 & c_1 & s_1 s_2 \\ 1 & 0 & c_2 \end{bmatrix} 8',
\end{aligned}$$

3.



4.



from the base coordinate to the end effector. (15')

$${}^0T_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad z'$$

$${}^1T_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad z'$$

$${}^0T_2 = \begin{bmatrix} \sin\theta_1 \cos\theta_1 & 0 & 0 \\ -\cos\theta_1 \sin\theta_1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad z$$

$${}^2T_3 = \begin{bmatrix} 1 & 0 & 0 & -l_1 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad z'$$

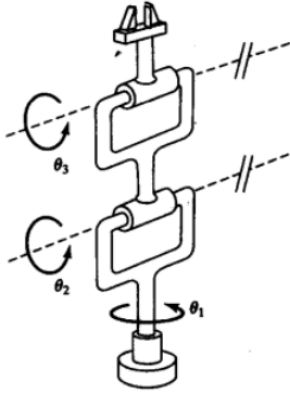
$${}^0T_3 = \begin{bmatrix} \sin\theta_1 & 0 & \cos\theta_1 & d_3 \cos\theta_1 - l_1 \sin\theta_1 \\ -\cos\theta_1 & 0 & \sin\theta_1 & d_3 \sin\theta_1 + l_1 \cos\theta_1 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad z''$$

$$\theta = \begin{bmatrix} \theta_1 \\ d_2 \\ d_3 \end{bmatrix} \quad z' \quad {}^0\vec{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^0\vec{z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^0\vec{z}_3 = \begin{bmatrix} \cos\theta_1 \\ \sin\theta_1 \\ 0 \end{bmatrix} \quad z'$$

$$RPP: \quad \xi_1 = 1 \quad \xi_2 = 0 \quad \xi_3 = 0 \quad z' \quad \vec{p} = \begin{bmatrix} d_3 \cos\theta_1 - l_1 \sin\theta_1 \\ d_3 \sin\theta_1 + l_1 \cos\theta_1 \\ d_2 \end{bmatrix} \quad z'$$

$$J(\vec{\theta}) = \begin{bmatrix} \frac{\partial p}{\partial \theta_1} & \frac{\partial p}{\partial d_2} & \frac{\partial p}{\partial d_3} \\ \xi_1 {}^0\vec{z}_1 & \xi_2 {}^0\vec{z}_2 & \xi_3 {}^0\vec{z}_3 \end{bmatrix} \quad 4'$$

$$J(\vec{\theta}) = \begin{bmatrix} -d_3 \sin\theta_1 - l_1 \cos\theta_1 & 0 & \cos\theta_1 \\ d_3 \cos\theta_1 - l_1 \sin\theta_1 & 0 & \sin\theta_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad 6'$$



α_{i-1}	a_{i-1}	d_i	θ_i
0	0	0	θ_1
90	0	0	θ_2
0	l_3	0	θ_3
-90	l_4	0	0

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{3'}$$

$${}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{3'}$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & l_3 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{3'}$$

$${}^3T_e = \begin{bmatrix} 1 & 0 & 0 & l_4 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{1'}$$

$${}^0T_e = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_e {}^{2'}$$

=

$$\begin{bmatrix} c_1c_2s_3 - c_1s_2s_3 & -c_1c_2s_3 - c_1s_2c_3 & s_1 & c_1c_2l_3 + l_4(c_1c_2s_3 - c_1s_2s_3) \\ s_1c_2c_3 - s_1s_2s_3 & -s_1c_2s_3 - s_1s_2c_3 & -c_1 & s_1c_2l_3 + l_4(s_1c_2c_3 - s_1s_2s_3) \\ s_2c_3 + c_2s_3 & -s_2s_3 + c_2c_3 & 0 & s_2l_3 + l_4(s_2c_3 + c_2s_3) \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{3'}$$

$$P = \begin{bmatrix} c_1 c_2 l_3 + l_4 (c_1 c_2 s_3 - c_1 s_2 s_3) \\ s_1 c_2 l_3 + l_4 (s_1 c_2 c_3 - s_1 s_2 s_3) \\ s_2 l_3 + l_4 (s_2 c_3 + c_2 s_3) \end{bmatrix} 2'$$

$$Z_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, Z_2 = \begin{pmatrix} s_1 \\ -c_1 \\ 0 \end{pmatrix}, Z_3 = \begin{pmatrix} s_1 \\ -c_1 \\ 0 \end{pmatrix} 3'$$

$$\xi_1 = 1, \xi_2 = 1, \xi_3 = 1, \quad 3'$$

$$\theta = [\theta_1 \ \theta_2 \ \theta_3]^T$$

$$J(\theta) = \begin{bmatrix} \frac{\partial P_x}{\partial \theta_1} & \frac{\partial P_x}{\partial \theta_2} & \frac{\partial P_x}{\partial \theta_3} \\ \frac{\partial P_y}{\partial \theta_1} & \frac{\partial P_y}{\partial \theta_2} & \frac{\partial P_y}{\partial \theta_3} \\ \frac{\partial P_z}{\partial \theta_1} & \frac{\partial P_z}{\partial \theta_2} & \frac{\partial P_z}{\partial \theta_3} \\ \xi_1 Z_1 & \xi_2 Z_2 & \xi_3 Z_3 \end{bmatrix} 4'$$

$$= \begin{bmatrix} -s_1 c_2 l_3 + l_4 (s_1 s_2 s_3 - s_1 c_2 s_3) & -c_1 s_2 l_3 - l_4 (c_1 s_2 s_3 + c_1 c_2 s_3) & l_4 (c_1 c_2 - c_1 s_2) c_3 \\ c_1 c_2 l_3 + l_4 (c_1 c_2 c_3 - c_1 s_2 s_3) & -s_1 s_2 l_3 - l_4 (s_1 s_2 c_3 + s_1 c_2 s_3) & l_4 (-s_1 c_2 s_3 - s_1 s_2 c_3) \\ 0 & c_2 l_3 + l_4 (c_2 c_3 - s_2 s_3) & l_4 (-s_2 s_3 + c_2 c_3) \\ 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 0 & 0 & 0 \end{bmatrix} 8'$$